12TH CHEMISTRY · CHAPTER 18 · SOLVED EXERCISE

Transition Metals

Complete textbook exercise (MCQs, Short Answer, Constructed Response, Descriptive) with Quick Check solutions

Q1. Multiple Choice Questions (MCQs)

I. Which statement about the general physical properties of the first-row transition elements (titanium to copper) is incorrect?

(a) They generally have high densities.
(b) They are typically soft solids at room temperature.
(c) They are good conductors of heat and electricity.
(d) They possess high melting and boiling points.

Answer: (b) They are typically soft solids at room temperature. — Transition metals exist as hard and rigid solids, not soft solids.

II. The ability of transition elements to behave as catalysts is primarily linked to which of their properties:

(a) Their high density.
(b) Their ability to easily change oxidation states.
(c) Their shiny surface appearance.
(d) Their formation of complex ions.

Answer: (b) Their ability to easily change oxidation states. — They switch easily between stable oxidation states, forming intermediates with lower activation energy.

III. What is the reason for variable oxidation states in 3d transition metals?

(a) High density.
(b) Stable, filled 4s subshell.
(c) Comparable energy of 3d and 4s subshells.
(d) High ionization energies.

Answer: (c) Comparable energy of 3d and 4s subshells. — Because 3d and 4s are very close in energy, electrons of both are used in bonding.

IV. Which ionic configuration represents a metal that lost only its 4s electrons?

(a) Ti^{4+} ([Ar])
(b) V^{5+} ([Ar])
(c) Fe^{2+} ([Ar] $3d^6$)
(d) Sc^{3+} ([Ar])

Answer: (c) Fe^{2+} ([Ar] $3d^6$) — Fe is [Ar] $3d^6$ $4s^2$; losing only the two 4s electrons gives Fe^{2+}, [Ar] $3d^6$. The others have lost 3d electrons as well.

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V. In the complex ion, $[Fe(CN)_6]^{4+}$, what is coordination number of central metal ion?

(a) 2 (b) 4
(c) 6 (d) 8

Answer: (c) 6 — Six CN^- ligands form six dative bonds with the iron ion.

VI. The colour displayed by a transition metal complex is a result of electrons moving between d-orbitals. This process is called a:

(a) p-p transition (b) d-d transition
(c) s-p transition (d) Ligand-to-metal charge transfer

Answer: (b) d-d transition — An electron is promoted from a lower-energy d-orbital to a higher-energy d-orbital by absorbing light.

VII. Adding excess ammonia to pink $[Co(H_2O)_6]^{2+}$ gives a yellow solution. What other changes occur besides colour change?

(a) Coordination number changes from 6 to 4
(b) Oxidation state of Co changes form +2 to +3
(c) d-orbital splitting magnitude (ΔE) changes
(d) Complex becomes a precipitate

Answer: (c) d-orbital splitting magnitude (ΔE) changes — $[Co(NH_3)_6]^{2+}$ is still octahedral cobalt(II) in solution; only the ligand, and therefore ΔE, has changed.

VIII. Which one of the following complex has different geometry than others?

(a) $[CuCl_4]^{2-}$
(b) $[Cu(H_2O)_4(OH)_2]$
(c) $[Cu(H_2O)_6]^{2+}$
(d) $[Cu(NH_3)_4(H_2O)_2]^{2+}$

Answer: (a) $[CuCl_4]^{2-}$ — $[CuCl_4]^{2-}$ is tetrahedral (four ligands); the other three have six ligands and are octahedral.

IX. What geometry is typical when $Co^{2+}_{(aq)}$ reacts with six monodentate anionic ligands?

(a) Octahedral (b) Square Planar
(c) Linear (d) Tetrahedral

Answer: (d) Tetrahedral — The textbook states that aqueous cobalt(II) ions usually form tetrahedral complexes with monodentate anionic ligands such as Cl^-, OH^- and SCN^-, e.g., $[CoCl_4]^{2-}$.

X. Optical isomers are characterized by their ability to rotate plane-polarized light. The isomer that rotates light to the right is called:

(a) Laevorotatory (–) (b) Superimposable
(c) Dextrorotatory (+) (d) Achiral

Answer: (c) Dextrorotatory (+) — Rotation to the right is dextrorotatory (+); to the left is laevorotatory (–).

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MCQ V: the ion is printed as $[Fe(CN)_6]^{4+}$; the correct charge is 4– (hexacyanoferrate(II)). The answer is not affected. MCQ IX: the stem says "six" ligands, but the tetrahedral complexes named in the textbook have four ligands; (d) is the answer according to the textbook statement.

Q2. SHORT ANSWER QUESTIONS

a) Why is zinc not included in the transition metals, although it shows electronic configuration with d-orbitals and also the member of d-block?

A transition element must have partially filled d-orbitals in at least one of its stable oxidation states. Zinc forms only one ion, Zn^{2+}, whose configuration is [Ar] $3d^{10}$ $4s^0$. Its 3d subshell is completely filled, so zinc is not classified as a transition element.

b) What is the correlation between the ability of transition elements to exhibit variable oxidation states and their electron configuration.

First row transition elements show variable oxidation states because the **3d and 4s subshells have comparable (very similar) energy**. The energy difference between the inner 3d subshell and the outermost 4s subshell is very small. So, like the 4s electrons, the 3d electrons can also easily be lost: both take part in chemical bonding. A 3d metal atom can lose:

- only the 4s electrons, usually giving the common **+2** oxidation state;
- the 4s electrons plus a variable number of 3d electrons, giving the **higher** oxidation states.

Among these metals Mn has the maximum oxidation state of +7. The oxidation state varies in a parabolic trend from left to right within the d-block.

c) How do transition metals function as effective catalysts. Give some examples.

1. Variable oxidation states (usually homogeneous catalysis): Transition elements can easily switch between their stable oxidation states (e.g., $Fe^{2+} \rightleftharpoons Fe^{3+}$). The metal ion forms an unstable intermediate compound with a reactant molecule. The intermediate quickly reacts to form the final product and regenerates the catalyst in its original oxidation state, providing a pathway of lower activation energy. Example: nickel salts used in the preparation of acrylic acid from acetylene involve the oxidation states Ni(0) and Ni(II).

2. Vacant, energetically accessible d-orbitals (generally heterogeneous catalysis): Reactant molecules attach to the catalyst surface and form a temporary activated complex, which holds the reactants in the correct orientation and lowers the activation energy. After the reaction the products desorb and the catalyst is ready for the next cycle. Example: solid iron accelerates the reaction between nitrogen and hydrogen to form ammonia in the Haber process.

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d) Why the complex $[Cr(en)_3]^{3+}$ is more stable than $[Cr(H_2O)_6]^{3+}$?

en (1,2-diaminoethane) is a bidentate ligand: each molecule forms two dative bonds with the chromium ion, making a ring. Water is a monodentate ligand. Complexes with polydentate ligands have higher stability constants than those with monodentate ligands; this is the **chelating effect**. So $[Cr(en)_3]^{3+}$ is more stable.

e) Define a complex and the nature of the bond between its components.

A **metal complex** (coordination compound) is a molecule or ion that forms when a central metal atom or ion is surrounded by a number of electron donor species (molecules or ions). The bond between the metal and each ligand is a **coordinate covalent (dative) bond**: the ligand donates the electron pair (Lewis base) and the metal accepts it (Lewis acid).

f) What is the coordination number? Give this number for $[Fe(CN)_6]^{4+}$.

The coordination number is the total number of dative bonds formed between the ligands and the central metal atom or ion in a complex. In this complex the iron ion is bonded to six CN^- ligands, so the coordination number is **6**.

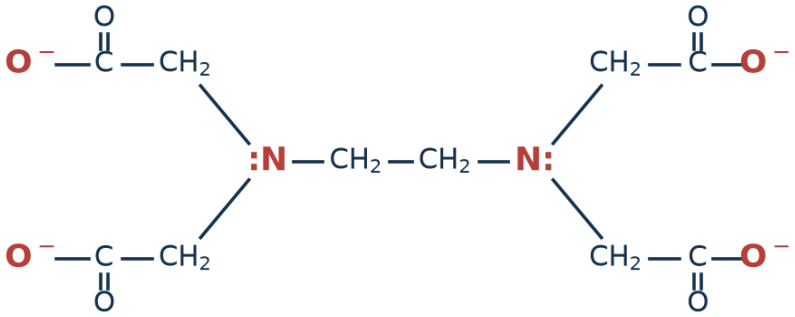
The charge is printed as 4+ in the question; the ion is $[Fe(CN)_6]^{4-}$.

g) Define a bidentate ligand and state how many bonds it forms to the central ion.

A bidentate ligand is a ligand that can donate two electron pairs to the central metal atom or ion. It forms **two** dative (coordinate covalent) bonds. Examples: 1,2-diaminoethane ($H_2NCH_2CH_2NH_2$, *en*) and the ethanedioate (oxalate) ion, $C_2O_4^{2-}$.

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h) Draw the structure of EDTA ligand and label its electron pair donating sites.



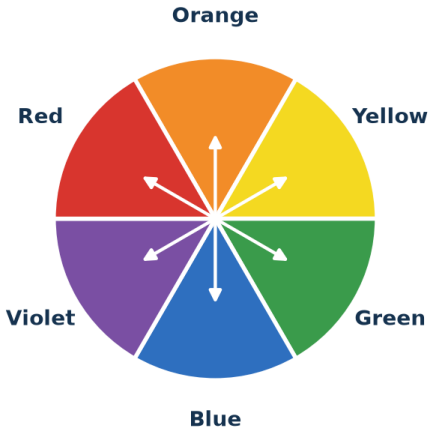
Electron pair donating sites (red): 2 nitrogen atoms + 4 oxygen atoms = 6

Ethylenediaminetetraacetate ion (EDTA)

EDTA is a hexadentate ligand: it attaches to a metal ion through six donor atoms (four oxygen atoms and two nitrogen atoms).

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i) Draw the colour wheel used to predict the colour of transition metal complexes. Also pair up the complementary colours with each other.



Colour wheel: opposite colours are complementary

Complementary pairs: Red ↔ Green, Orange ↔ Blue, Yellow ↔ Violet.

The colour seen is the complementary colour of the light absorbed; e.g., a complex that absorbs yellow light appears violet.

j) Define K_{stab}. How it reflects the stability of coordination complexes?

K_{stab} (stability constant) is the equilibrium constant for the formation of a complex ion in a solvent from its constituent ions or molecules. The higher the value of K_{stab} (or $\log_{10}K_{stab}$), the more stable the complex, and the equilibrium lies far to the right. A lower value means a less stable complex whose ligands can be easily displaced.

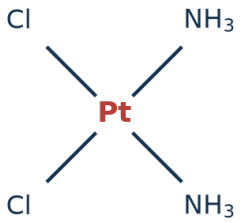
k) How coordination number 4 has two shapes i.e., tetrahedral and square planar?

With four ligands the shape depends on the electronic configuration of the metal and the nature of the ligands. In the **square planar** shape the metal ion and the four ligands lie in one plane at the corners of a square (bond angle 90°). It is characteristic of d^8 metal ions (Ni^{2+}, Pt^{2+}) with ligands like NH_3 and CN^-, e.g., $[Ni(CN)_4]^{2-}$. In the **tetrahedral** shape the four ligands point to the corners of a tetrahedron (109.5°). It is common for d^7, d^8 and d^{10} ions, e.g., $[CoCl_4]^{2-}$.

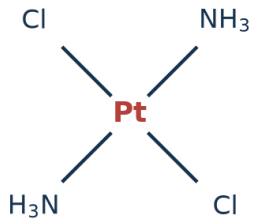
l) Illustrate the type of isomerism shown by $[PtNH_3)_2Cl_2]$ complex.

It shows **geometrical (cis-trans) isomerism**.

In *cis*-$[Pt(NH_3)_2Cl_2]$ (cisplatin) the two chlorine atoms are adjacent, at 90° in the square plane. In *trans*-$[Pt(NH_3)_2Cl_2]$ (transplatin) they are opposite, at 180°.



cis-platin (polar)



trans-platin (non-polar)

Geometrical isomers of $[Pt(NH_3)_2Cl_2]$

The properties of the two isomers are different. Cisplatin is used as an anti-cancer drug; transplatin does not show this property. In transplatin the bond dipoles of the identical ligands point in opposite directions and cancel, so the molecule is **non-polar**. In cisplatin they do not cancel, giving a non-zero dipole moment, so it is **polar**.

m) Give medicinal importance of cis-plantin.

Cisplatin, *cis*-diamminedichloroplatinum(II), is a landmark inorganic anticancer drug. Its discovery was accidental: platinum electrodes were found to inhibit bacterial cell division. It targets rapidly dividing cancer cells by entering the cell and undergoing ligand exchange, in which water replaces its chloride ions. The “activated” drug then binds to guanine bases in DNA, creating cross-links that link the double helix. This structural damage prevents DNA replication and triggers programmed cell death (apoptosis).

“cis-plantin” is printed in the textbook; the compound is cis-platin (cisplatin).

n) What happens when excess of ammonia solution is added to Co(II) hydroxide precipitates?

A ligand exchange reaction takes place. Ammonia ligands replace the water and hydroxide ligands, the blue precipitate dissolves, and a yellow solution of the soluble hexaamminecobalt(II) ion is formed.

$$[Co(H_2O)_4(OH)_2]_{(s)} + 6NH_{3(aq)} \rightarrow [Co(NH_3)_6]^{2+}{}_{(aq)} + 4H_2O_{(l)} + 2OH^-_{(aq)}$$

o) Is there any effect of excessive dilution of complexes? If yes then justify how does dilution affect the position of equilibrium in ligand ion exchange reactions?

Yes. If a stable complex is diluted with a massive amount of water, the equilibrium can be forced back to the left, re-forming the original hydrated ion. Water is itself a ligand; a very large excess of it competes with the other ligand and, by Le Chatelier’s principle, the position of equilibrium shifts towards the aqua complex.

Q3. CONSTRUCTED RESPONSE QUESTIONS

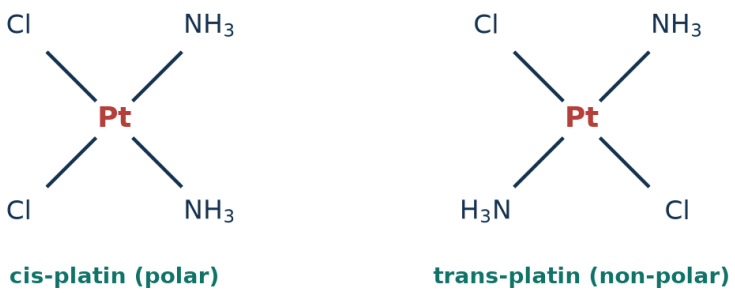
a) Discuss the two types of stereoisomerism found in transition metal complexes. Use the examples $[Pt(NH_3)_2Cl_2]$ (square planar) and $[Ni(NH_2CH_2CH_2NH_2)_3]^{2+}$ (octahedral) to illustrate the key structural differences that give rise to geometrical and optical isomerism, respectively.

Compounds that have the same chemical formula and bonds but differ in the arrangement of the ligands in space are called **stereoisomers**. There are two types: **geometrical (cis-trans) isomerism** and **optical isomerism**.

1. Geometrical (cis-trans) isomerism

In cis-trans isomers, identical ligands are present at either adjacent positions (90° angle, **cis**) or opposite positions (180° angle, **trans**). It is common in square planar complexes (e.g., $[Pt(NH_3)_2Cl_2]$) and octahedral complexes (e.g., $[Co(NH_3)_4Cl_2]$).

In *cis*-$[Pt(NH_3)_2Cl_2]$ (cisplatin) the two chlorine atoms are adjacent, at 90° in the square plane. In *trans*-$[Pt(NH_3)_2Cl_2]$ (transplatin) they are opposite, at 180°.

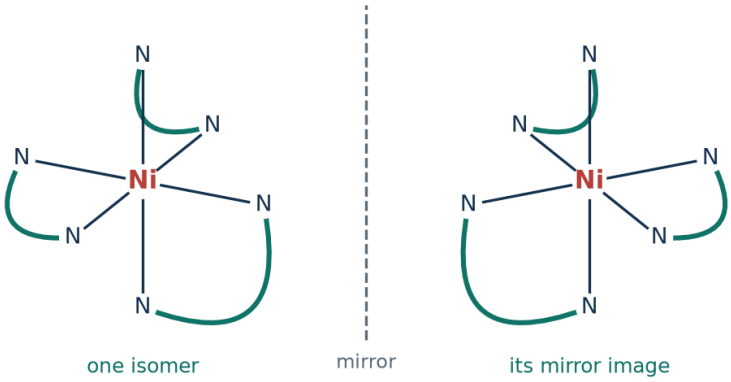


Geometrical isomers of $[Pt(NH_3)_2Cl_2]$

The properties of the two isomers are different. Cisplatin is used as an anti-cancer drug; transplatin does not show this property. In transplatin the bond dipoles of the identical ligands point in opposite directions and cancel, so the molecule is **non-polar**. In cisplatin they do not cancel, giving a non-zero dipole moment, so it is **polar**.

2. Optical isomerism

Optical isomers are non-superimposable mirror images of each other. Optical isomerism is commonly shown by octahedral complexes with bidentate ligands, e.g., the nickel(II) complex with 1,2-diaminoethane, $[Ni(en)_3]^{2+}$. Its two isomers are mirror images that cannot be superimposed; such pairs are called **enantiomers**. They rotate the plane of polarized light in opposite directions: the isomer that rotates it to the right is **dextrorotatory (+)**, the one that rotates it to the left is **laevorotatory (–)**.



The two non-superimposable optical isomers of $[Ni(en)_3]^{2+}$ (each curve is one en ligand)

Geometrical isomerism	Optical isomerism
Identical ligands are at adjacent (cis, 90°) or opposite (trans, 180°) positions.	The isomers are non-superimposable mirror images (enantiomers).
Common in square planar and octahedral complexes.	Common in octahedral complexes with bidentate ligands.
Isomers differ in polarity and other properties.	Isomers rotate plane-polarized light in opposite directions.

b) The reaction between Cu^{2+} and I^- results in a solid precipitate of CuI and aqueous I_2.

$$2Cu^{2+}_{(aq)} + 4I^-_{(aq)} \rightarrow 2CuI_{(s)} + I_{2(aq)}$$

a) Write down the two relevant reduction half-equations and their E° values (provided in the text).

$$\tfrac{1}{2}I_{2(aq)} + e^- \rightarrow I^-_{(aq)} \quad E° = +0.54\ V$$
$$Cu^{2+}_{(aq)} + e^- \rightarrow Cu^+_{(aq)} \quad E° = +0.15\ V$$

b) Explain why comparing only the E° values suggests the reaction is not feasible as written.

The I_2/I^- couple has the higher (more positive) E°, so iodine should be reduced rather than Cu^{2+}; Cu^+ is a better reducing agent than I^-. For the reaction as written, $E°_{cell} = E°_{reduction} - E°_{oxidation} = (+0.15) - (+0.54) = -0.39$ V. A negative value suggests the reaction is not feasible.

c) Use Le Chatelier's principle to explain why the reaction proceeds forward in practice.

The CuI formed is insoluble and precipitates continuously out of the solution. This removes the Cu^+ product, so by Le Chatelier's principle the equilibrium shifts to the right to replace it, and the reaction goes forward.

c) Using the examples $[Co(NH_3)_6]^{3+}$ and $[Fe(CN)_6]^{3-}$, demonstrate how to predict the formula and overall charge of a complex ion. Clearly explain the role of the central metal ion charge, ligand charge, and coordination number in determining the final result for both examples.

The general structure of a complex ion is $[M(L)_x]^{\pm y}$. The metal is written first, then the ligand with its number as subscript, all inside square brackets.

- **Central metal ion charge:** its oxidation state is the starting value of the charge.
- **Ligand charge:** each ligand adds its own charge (0 for neutral ligands, –1 for anionic ligands such as cyanide).
- **Coordination number:** gives the number of ligands (x) attached to the metal.

Charge of complex = metal charge + (number of ligands × ligand charge)

	$[Co(NH_3)_6]^{3+}$	$[Fe(CN)_6]^{3-}$
Metal ion and oxidation state	Co, +3	Fe, +3
Ligand, type and charge	NH_3 (ammine), neutral, 0	CN^- (cyano), anionic, –1
Coordination number	6	6
Formula	Co first, then (NH_3) with subscript 6, in [ ]	Fe first, then (CN) with subscript 6, in [ ]
Charge calculation	(+3) + [6 × (0)] = +3	(+3) + [6 × (–1)] = –3
Result	Cationic complex	Anionic complex

DESCRIPTIVE QUESTIONS

Q4. Explain the factors that determine the specific geometry (shape) of a transition metal complex.

The geometry (shape) of a transition metal complex depends on:

1. the **electronic configuration** of the metal;
2. the **coordination number** (number of donor atoms attached) of the metal;
3. the **nature of the ligands**.

The common geometries that result are:

Geometry	C.N.	Bond angle	Description	Examples
Linear	2	180°	Two ligands on opposite sides of the central metal ion	$[Ag(NH_3)_2]Cl$, $K[Ag(CN)_2]$

Geometry	C.N.	Bond angle	Description	Examples
Square planar	4	90°	Metal ion and four ligands in the same plane, at the corners of a square; characteristic of d^8 ions (Ni^{2+}, Pt^{2+}) with ligands like NH_3 and CN^-	$[Ni(CN)_4]^{2-}$, $[Pt(NH_3)_4]^{2+}$
Tetrahedral	4	109.5°	Four ligands at the corners of a tetrahedron; common when the metal ion has a d^7, d^8 or d^{10} configuration	$[CoCl_4]^{2-}$, $[MnCl_4]^{2-}$
Octahedral	6	90° (180° for opposite ligands)	Six ligands along the x, y and z axes: four in a square plane, one above and one below it. The most common geometry	$[Cr(NH_3)_6]^{3+}$, $[Co(en)_3]^{3+}$

Role of each factor: coordination number 2 gives a linear shape and 6 gives an octahedral shape. For coordination number 4, the electronic configuration and the ligand decide the shape: d^8 ions such as Ni^{2+} and Pt^{2+} with ligands like NH_3 and CN^- are square planar, while d^7, d^8 and d^{10} ions (e.g., with Cl^-) are tetrahedral.

Q3. Describe the splitting of d-orbitals and colour formation in octahedral and tetrahedral complexes.

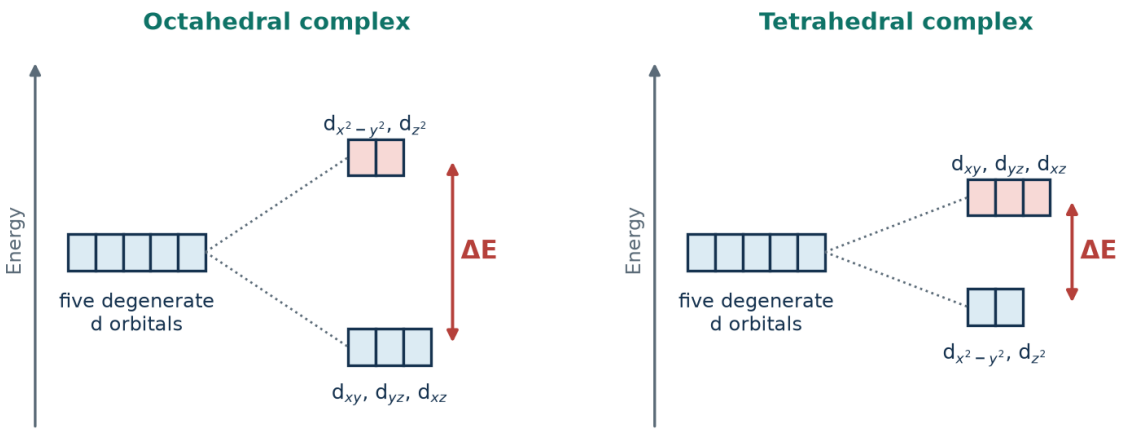
Splitting of d-orbitals

The d-orbitals have a characteristic cloverleaf shape, generally four lobes (two dumbbells in a plane). There are five of them: d_{xy}, d_{xz}, d_{yz}, $d_{x^2-y^2}$ and d_{z^2}. In an isolated atom all five have the same energy (**degenerate**). When ligands approach the metal ion to form a complex, they create an electric field that splits the five d-orbitals into two groups of different energy (**non-degenerate**). The energy difference between the two sets is represented by **Δ (ΔE)**.

Octahedral complexes (six ligands), e.g., $[Ti(H_2O)_6]^{3+}$: the d-orbitals split into three lower-energy orbitals (d_{xy}, d_{xz}, d_{yz}) and two higher-energy orbitals ($d_{x^2-y^2}$, d_{z^2}).

Tetrahedral complexes (four ligands), e.g., $[CuCl_4]^{2-}$: the splitting is inverted, with three higher-energy orbitals (d_{xy}, d_{xz}, d_{yz}) and two lower-energy orbitals ($d_{x^2-y^2}$, d_{z^2}). The magnitude of splitting is smaller than in octahedral complexes.

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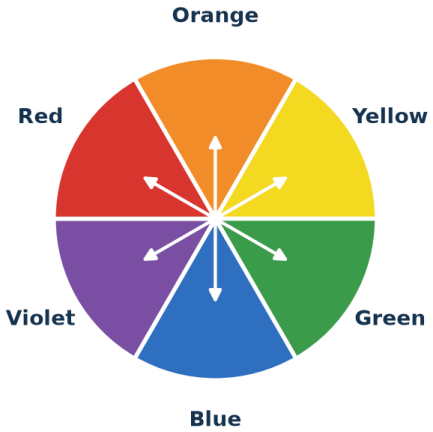
Splitting of the d-orbitals in octahedral and tetrahedral fields

Colour formation

When white light falls on a compound: if all of it is absorbed, the complex appears black; if all is reflected or transmitted, it appears white; if part is absorbed, the complex shows the colour of the remaining light. The colour we see is the **complementary colour** (opposite on the colour wheel) of the light absorbed. For example, a complex that absorbs yellow light appears violet.

Electrons in a lower-energy d-orbital absorb light and are promoted (excited) to a higher-energy d-orbital. This is called a **d-d transition**. The unabsorbed light is transmitted and decides the colour. The energy needed corresponds to a specific frequency (ν) and wavelength (λ) of visible light by Planck's equation:

$$\Delta E = h\nu = hc/\lambda \text{ (h = Planck's constant, c = speed of light)}$$



Colour wheel: opposite colours are complementary

Complementary pairs: Red ↔ Green, Orange ↔ Blue, Yellow ↔ Violet.

Effect of different ligands on ΔE and colour

The nature of the ligand affects the magnitude of the splitting, ΔE. Ligands like NH_3 and CN^- cause larger splitting; for ligands like Cl^- and I^-, ΔE is smaller. A larger ΔE means light of higher frequency (shorter wavelength), in the violet or blue region, is absorbed, so the compound reflects yellow light and looks yellow. Complexes that absorb low-energy light (red/orange) appear blue or green. Order of ligands by their ability to split the d-orbitals (high ΔE to low ΔE):

$$CN^- > en > NH_3 > H_2O > OH^- > Cl^- > Br^- > I^-$$

Q4. Compare the reactions of conc. HCl with aqueous Cu^{2+} and aqueous Co^{2+}. Include equations, colour/geometry changes, and the reason for different colours.

$$[Cu(H_2O)_6]^{2+}_{(aq)} + 4Cl^-_{(aq)} \rightarrow [CuCl_4]^{2-} + 6H_2O$$
$$[Co(H_2O)_6]^{2+}_{(aq)} + 4Cl^-_{(aq)} \rightarrow [CoCl_4]^{2-}_{(aq)} + 6H_2O_{(l)}$$

	Aqueous Cu^{2+}	Aqueous Co^{2+}
Starting complex	$[Cu(H_2O)_6]^{2+}$	$[Co(H_2O)_6]^{2+}$
Initial colour	Blue	Pink
Product	$[CuCl_4]^{2-}$	$[CoCl_4]^{2-}$
Final colour	Yellow (yellow-green)	Blue
Geometry change	Octahedral → tetrahedral	Octahedral → tetrahedral
Coordination number	6 → 4	6 → 4

In both cases conc. HCl is added drop by drop and four chloride ions replace six water ligands.

Reason for the different colours: the exchange of ligands (H_2O by Cl^-) changes the magnitude of d-orbital splitting, ΔE, so the frequency of light absorbed changes and a new complementary colour is seen. The two final colours differ from each other because the metal ions are different: Cu^{2+} ($3d^9$) and Co^{2+} ($3d^7$) have different numbers of d electrons and different ΔE values, so they absorb light of different frequencies.

Q5. Describe the various redox reactions that takes place in transition metal complex ions?

Transition elements can exist in multiple oxidation states. The feasibility of a redox reaction is found by comparing the standard electrode potentials (E°) of the half-reactions. A reaction is feasible if the overall cell potential is **positive**.

$$E°_{cell} = E°_{reduction} - E°_{oxidation}$$

The species with the more positive E° value is reduced (gains electrons); the species with the more negative (less positive) value is oxidized (loses electrons).

1. Acidified manganate(VII) ions with ethanedioate ions

The half-reaction with the more positive E° is more likely to undergo reduction. So MnO_4^- is reduced to Mn^{2+} (it is the oxidizing agent) and $C_2O_4^{2-}$ is oxidized to CO_2 (it is the reducing agent).

$$2CO_{2(g)} + 2e^- \rightarrow C_2O_4^{2-}{}_{(aq)} \quad E° = +0.49\ V$$
$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O_{(l)} \quad E° = +1.52\ V$$

The electrons lost must equal the electrons gained, so the reduction half-equation is multiplied by 2 and the oxidation half-equation by 5:

$$2 \times [MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O_{(l)}]$$
$$5 \times [C_2O_4^{2-}{}_{(aq)} \rightarrow 2CO_{2(g)} + 2e^-]$$
$$2MnO_4^- + 5C_2O_4^{2-}{}_{(aq)} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_{2(g)} + 8H_2O_{(l)}$$
$$E°_{cell} = (+1.52) - (+0.49) = +1.03\ V$$

The large positive $E°_{cell}$ (+1.03 V) shows that the forward reaction is feasible.

The textbook prints the overall equation as "$2MnO_4^- + 5C_2O_4^{2-} + 8H^+ \rightarrow 5Mn^{2+} + 2CO_2 + 8H_2O$", which is not balanced. The balanced equation is given above; $E°_{cell} = +1.03$ V is as in the textbook.

2. Manganate(VII) ion with iron(II) ion

MnO_4^- is reduced and Fe^{2+} is oxidized, as indicated by the potential values.

$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O_{(l)} \quad E° = +1.52\ V$$
$$Fe^{3+} + e^- \rightarrow Fe^{2+} \quad E° = +0.77\ V$$
$$5 \times [Fe^{2+} \rightarrow Fe^{3+} + e^-]$$
$$MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O_{(l)}$$
$$E°_{cell} = (+1.52) - (+0.77) = +0.75\ V$$

The positive value shows that the reaction proceeds to the right, i.e., it is feasible.

The textbook adds the two values and prints E° = +2.29 V (and writes the product as $5Fe^{2+}$). Using the textbook's own formula $E°_{cell} = E°_{reduction} - E°_{oxidation}$, the value is +1.52 – 0.77 = +0.75 V. Either way the value is positive and the reaction is feasible.

3. Copper(II) ion and iodide ion

When an excess of potassium iodide is added to aqueous copper(II) sulphate, a white precipitate of copper(I) iodide forms in a red-brown solution of iodine:

$$2Cu^{2+}{}_{(aq)} + 4I^-{}_{(aq)} \rightarrow 2CuI_{(s)} + I_{2(aq)}$$
$$\tfrac{1}{2}I_{2(aq)} + e^- \rightarrow I^-{}_{(aq)} \quad E° = +0.54\ V$$
$$Cu^{2+}{}_{(aq)} + e^- \rightarrow Cu^+{}_{(aq)} \quad E° = +0.15\ V$$

The I_2/I^- couple has the higher E°, so it would be more likely to undergo reduction; the Cu^{2+}/Cu^+ couple has the lower E°, suggesting Cu^+ is a better reducing agent than I^-. So

the calculated net standard potential is **negative** (0.15 – 0.54 = –0.39 V) and the reaction looks not feasible.

However, the reaction does proceed forward. CuI is continuously precipitated out of the solution, which removes the Cu^{+} product and shifts the equilibrium to the right (**Le Chatelier's principle**). The amount of I_2 formed can be found by titrating with a thiosulphate solution.

Q6. Explain the stereoisomerism with its types in transition metal complexes.

Compounds that have the same chemical formula and bonds but differ in the arrangement of the ligands in space are called **stereoisomers**. There are two types: **geometrical (cis-trans) isomerism** and **optical isomerism**.

1. Cis-trans (geometrical) isomerism

In cis-trans isomers, identical ligands are present at either adjacent positions (90° angle, **cis**) or opposite positions (180° angle, **trans**). It is common in square planar complexes (e.g., $[Pt(NH_3)_2Cl_2]$) and octahedral complexes (e.g., $[Co(NH_3)_4Cl_2]$).

Square planar example: cisplatin and transplatin, $[Pt(NH_3)_2Cl_2]$ (structures drawn in Q_2 I). In cisplatin the Cl atoms are at 90° and the molecule is polar; in transplatin they are at 180° and the molecule is non-polar. Cisplatin is an anti-cancer drug; transplatin is not.

Octahedral example:

$[Co(NH_3)_4(H_2O)_2]^{3+}$ is an octahedral complex of the type $[MA_4B_2]$. In the cis isomer the two H_2O ligands occupy adjacent positions at 90° to each other; in the trans isomer they occupy opposite positions at 180°.

2. Optical isomerism

Optical isomers are non-superimposable mirror images of each other. Optical isomerism is commonly shown by octahedral complexes with bidentate ligands, e.g., the nickel(II) complex with 1,2-diaminoethane, $[Ni(en)_3]^{2+}$. Its two isomers are mirror images that cannot be superimposed; such pairs are called **enantiomers**. They rotate the plane of polarized light in opposite directions: the isomer that rotates it to the right is **dextrorotatory (+)**, the one that rotates it to the left is **laevorotatory (–)**.

The structures of the two optical isomers are drawn in Q3 (a).

The descriptive questions are numbered Q4, Q3, Q4, Q5, Q6 in the textbook; the printed numbers are kept here.

QUICK CHECK QUESTIONS (IN-CHAPTER)

These questions appear in the "Quick Check" boxes inside the chapter, not in the end-of-chapter exercise.

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Quick Check 18.1

a) Write down electronic configuration of Co^{2+} and Co^{3+}, and find the number of unpaired electrons in d-orbitals.

Co^{2+}: [Ar] $3d^7$ $4s^0$, with **3** unpaired d electrons. Co^{3+}: [Ar] $3d^6$ $4s^0$, with **4** unpaired d electrons (the five d-orbitals are first filled singly, then pairing starts).

b) Why are the density and melting points of transition metals much higher than Group 1 metals?

Transition metal atoms are smaller and heavier, with a higher nuclear charge, so more mass is packed into a small volume (high density). Both their 4s and 3d electrons take part in metallic bonding, while a Group 1 metal gives only one electron. The metallic bonds are therefore much stronger and need more heat to break (high melting point).

c) The Sc^{3+} ion is [Ar] $3d^0$ $4s^0$. Why is scandium not classified as a transition element?

A transition element must have a partially filled d-subshell in a stable oxidation state. Scandium forms only Sc^{3+}, whose 3d subshell is empty ($3d^0$), so it is not classified as a transition element.

Quick Check 18.2

a) How Vanadium can show the range of oxidation states from +2 to +5?

Vanadium is [Ar] $3d^3$ $4s^2$. Its 3d and 4s subshells have very similar energy. Losing the two 4s electrons gives +2; losing in addition one, two or three 3d electrons gives +3, +4 and +5.

b) What is the role of a ligand in forming a complex ion, and what type of bond is formed?

The ligand acts as a Lewis base: it donates a lone pair of electrons into a vacant d-orbital of the central metal atom or ion. The bond formed is a coordinate covalent (dative) bond.

c) Why transition metals use both 4s and 3d electrons in bonding, unlike s-block elements.

In transition metals the energy difference between the inner 3d subshell and the outer 4s subshell is very small, so the 3d electrons can be lost as easily as the 4s electrons. In s-block elements the electrons below the outer s-subshell belong to a stable noble-gas core of much lower energy, so only the s electrons are used.

d) Write the electronic configuration for Fe^{3+}. Why is the Fe^{2+}/Fe^{3+} change useful in catalysis?

Fe^{3+}: [Ar] $3d^5$ $4s^0$. Iron switches easily between Fe^{2+} and Fe^{3+}. So the ion can form an unstable intermediate with a reactant and then be regenerated in its original oxidation state, giving a reaction pathway of lower activation energy.

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e) Which properties of transition metals enable them to be used as heterogeneous catalysts and homogeneous catalysts.

Heterogeneous catalysts: vacant, energetically accessible d-orbitals, which let reactant molecules attach to the metal surface (e.g., iron in the Haber process). **Homogeneous catalysts:** more than one stable oxidation state, between which the metal ion can easily switch (e.g., Ni(0)/Ni(II)).

Quick Check 18.3

a) How Lewis acid base concept explains the transition metal complex formation?

The central transition metal atom or ion is a Lewis acid: it accepts electron pairs into its vacant d-orbitals. Each ligand is a Lewis base: it donates a lone pair. This donation and acceptance forms coordinate covalent bonds and gives a stable complex.

b) Determine the overall charge of a complex ion formed from a Cr^{3+} metal ion with six neutral H_2O ligands and a coordination number of 6. Show your calculation.

Charge = metal charge + (number of ligands × ligand charge) = (+3) + [6 × (0)] = **+3**. The complex ion is $[Cr(H_2O)_6]^{3+}$.

c) Predict the likely geometry for the complex ion $[MnCl_4]^{2-}$, given that the central metal ion has a d^5 electron configuration and a coordination number of 4.

Tetrahedral, with bond angles of 109.5°. Square planar geometry is characteristic of d^8 ions, and the textbook lists $[MnCl_4]^{2-}$ as a tetrahedral complex.

d) EDTA is described as a hexadentate ligand. Explain what "hexadentate" means in terms of coordinate covalent bonds formed with a single metal ion.

Hexadentate means that one EDTA ion forms **six** coordinate covalent bonds with a single metal ion, through six donor atoms (four oxygen atoms and two nitrogen atoms).

e) Identify the two essential features of a transition metal ion that enable it to form complex ions.

(i) It has vacant (partially filled or empty) d-orbitals. (ii) These orbitals are energetically accessible, i.e., low enough in energy to accept the lone pairs donated by ligands.

Quick Check 18.4

a) How does the presence of ligands affect the degeneracy of d-orbitals in a complexion?

In an isolated atom the five d-orbitals are degenerate. The approaching ligands create an electric field that splits them into two sets of different energy, so they become non-degenerate. The energy gap between the sets is ΔE.

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b) Describe the key difference in d-orbital splitting patterns between octahedral complexes and tetrahedral complexes.

In octahedral complexes three orbitals (d_{xy}, d_{xz}, d_{yz}) are lower and two ($d_{x^2-y^2}$, d_{z^2}) are higher in energy. In tetrahedral complexes the pattern is inverted (three higher, two lower) and the splitting is smaller.

c) Explain the relationship between the absorbed light energy (ΔE) and the observed colour of a transition metal complex. Reference Planck's equation (ΔE=hν=hc/λ).

The complex absorbs light whose energy equals ΔE; by $\Delta E = h\nu = hc/\lambda$ this fixes its frequency and wavelength. A larger ΔE means higher frequency (shorter wavelength) is absorbed. The remaining light is transmitted, so the observed colour is the complementary colour of the light absorbed.

d) If a complex containing the ligand Cl^- absorbs light in the red/orange region of the spectrum, what general colour range would you expect the complex to appear?

It would appear **blue or green**, the complementary colours of orange and red.

e) Explain the phenomenon (d-d transitions) responsible for the colour of transition metal compounds.

An electron in a lower-energy d-orbital absorbs light from the visible region and is promoted (excited) to a higher-energy d-orbital. This is a d-d transition. The unabsorbed light is transmitted and gives the compound its colour.

Quick Check 18.5

a) Write the equation for the reaction, where $[Cu(H_2O)_4(OH)_2]_{(s)}$ dissolves in excess ammonia.

$$[Cu(H_2O)_4(OH)_2]_{(s)} + 4NH_{3(aq)} \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+}{}_{(aq)} + 2H_2O_{(l)} + 2OH^-_{(aq)}$$

The light blue precipitate dissolves to give a dark blue solution.

b) Pink $[Co(H_2O)_6]^{2+}$ turns blue with conc. HCl. Explain this colour change using d-orbital splitting.

The Cl^- ligands replace the water ligands and the complex changes from octahedral $[Co(H_2O)_6]^{2+}$ to tetrahedral $[CoCl_4]^{2-}$. Cl^- causes less splitting than H_2O, and tetrahedral splitting is smaller than octahedral, so ΔE decreases. Light of lower frequency is now absorbed, and the complementary colour seen changes from pink to blue.

c) For the reaction: $[Co(H_2O)_6]^{2+} + 4Cl^- \rightarrow [CoCl_4]^{2-} + 6H_2O$, identify the coordination number and geometry of the product complex.

In $[CoCl_4]^{2-}$ the coordination number is **4** and the geometry is **tetrahedral**.

Quick Check 18.6

a) Write the K_{stab} equilibrium expression for the formation of $[Cu(NH_3)_4]^{2+}$ from $[Cu(H_2O)_6]^{2+}$ and NH_3.

$$[Cu(H_2O)_6]^{2+} + 4NH_3 \rightleftharpoons [Cu(NH_3)_4]^{2+} + 6H_2O$$

$$K_{stab} = [[Cu(NH_3)_4]^{2+}] / ([[Cu(H_2O)_6]^{2+}] [NH_3]^4)$$

$[H_2O]$ is not included.

b) Explain why polydentate ligands form more stable complexes using K_{stab} data ($\log_{10}K_{stab}$) data for $[CuCl_4]^{2-}$ (5.6) and $[Cu(EDTA)]^{2-}$ (18.8).

$\log_{10}K_{stab}$ of $[Cu(EDTA)]^{2-}$ (18.8) is far higher than that of $[CuCl_4]^{2-}$ (5.6), so its equilibrium lies much further to the right. Cl^- is monodentate, while one EDTA ion holds the metal through six dative bonds. This extra stability of complexes with polydentate ligands is the chelating effect.

c) Write the balanced equation for the first step of the stepwise substitution of water ligands by ammonia in $[Cu(H_2O)_6]^{2+}$.

$$[Cu(H_2O)_6]^{2+}_{(aq)} + NH_{3(aq)} \rightleftharpoons [Cu(NH_3)(H_2O)_5]^{2+}_{(aq)} + H_2O_{(l)}$$

Quick Check 18.7

a) Which type of geometries shows cis-trans isomerism?

Square planar and octahedral geometries.

b) What type of ligand is most often required for a complex to show optical isomerism?

A bidentate ligand (in an octahedral complex), e.g., 1,2-diaminoethane.

c) For the complex $[Ni(en)_3]^{2+}$, state the coordination number and the type of ligand.

Coordination number = **6** (three *en* ligands, each forming two dative bonds). *en* (1,2-diaminoethane) is a neutral **bidentate** ligand.

d) Identify the type of isomerism displayed by the complex $[Ni(NH_2CH_2CH_2NH_2)_3]^{2+}$, and describe how the two different isomers interact with plane-polarized light.

It shows **optical isomerism**. The two isomers rotate the plane of polarized light in opposite directions: one to the right (dextrorotatory, +) and the other to the left (laevorotatory, –).

e) Define optical isomers (enantiomers) in terms of their mirror images and superimposability.

Optical isomers (enantiomers) are a pair of molecules that are mirror images of each other and cannot be superimposed on each other.

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Quick Check 18.8

a) State the principle used to determine the feasibility of a redox reaction using standard electrode potentials (E°), as described in the text.

The E° values of the two half-reactions are compared. A reaction is feasible if the overall cell potential, $E°_{cell} = E°_{reduction} – E°_{oxidation}$, is positive. The species with the more positive E° is reduced; the one with the less positive E° is oxidized.

b) For the reaction between MnO_4^- and Fe^{2+}, the relevant reduction potentials are MnO_4^-/Mn^{2+} (E° = +1.52 V) and Fe^{3+}/Fe^{2+} (E° = +0.77 V). Identify the oxidizing agent and the reducing agent in this specific feasible reaction.

MnO_4^- has the more positive E°, so it is reduced: it is the **oxidizing agent**. Fe^{2+} is oxidized to Fe^{3+}: it is the **reducing agent**.

c) Explain why water is omitted when calculating the $E°_{cell}$ for the reaction between MnO_4^- and $C_2O_4^{2-}$ ions, even though it appears in the overall balanced equation.

$E°_{cell}$ is calculated only from the E° values of the two half-reactions (+1.52 V and +0.49 V). Water is the solvent: it is neither oxidized nor reduced and has no electrode potential of its own in this reaction, so it does not enter the calculation.

The textbook does not state this reason directly; the answer follows from the method it uses.

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